Belady's Anomaly

- If number of frames increases, fewer page faults should occur. But, sometimes the reverse happens, i.e. more page faults occur when more frames are allocated to a process. This most unexpected result is termed **Belady's Anomaly**.

- So, increasing the number of page frames results in an increase in the number of page faults for certain memory access patterns.

- Belady's anomaly **occurs only in case of FIFO**, for certain set of reference string.

- It **never occurs for LRU and Optimal** page replacement. Because of **Stack Property.**

- According to stack property: At each stage, the **set of pages that were present in the main memory when number of frames is 'm' will be compulsorily present in the corresponding stages in main memory when the number of frames is increased to 'm+1'.**

- LRU and Optimal are stack based algorithm.

Least Recently Used (LRU)

Reference String:

0	1	2	3	0	1	4	0*	1*	2	3	4
0	0	0	3	3	3	4			2	2	2
	1	1	1	0	0	0			0	3	3
		2	2	2	1	1			1	1	4

No of Page Faults: 10
No of frames 3

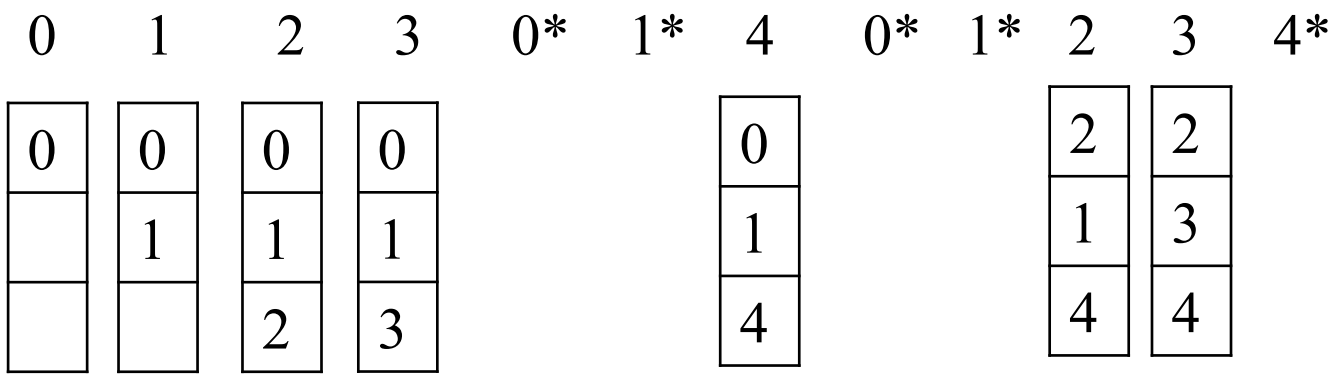
Reference String:

0	1	2	3	0*	1*	4	0*	1*	2	3	4
0	0	0	0			0			0	0	4
	1	1	1			1			1	1	1
		2	2			4			4	3	3
			3			3			2	2	2

No of Page Faults: 8
No of frames 4

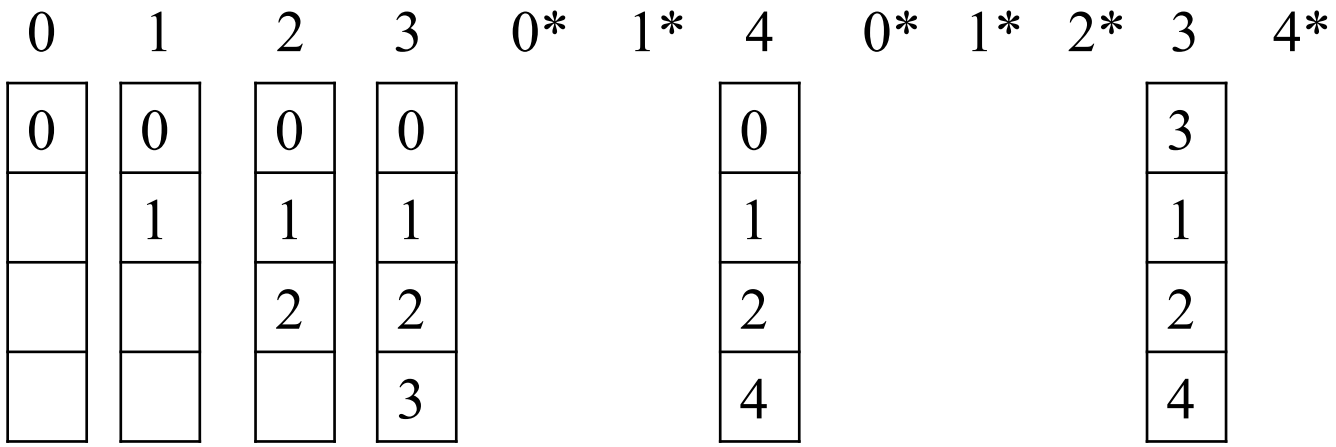
Optimal Page Replacement (Opt)

Reference String:



No of Page Faults: 7
No of frames 3

Reference String:



No of Page Faults: 6
No of frames 4

First In First Out (FIFO)

Reference String:

0	1	2	3	0	1	4	0*	1*	2	3	4*
0	0	0	3	3	3	4			4	4	
	1	1	1	0	0	0			2	2	
		2	2	2	1	1			1	3	

No of Page Faults: 9
No of frames 3

Reference String:

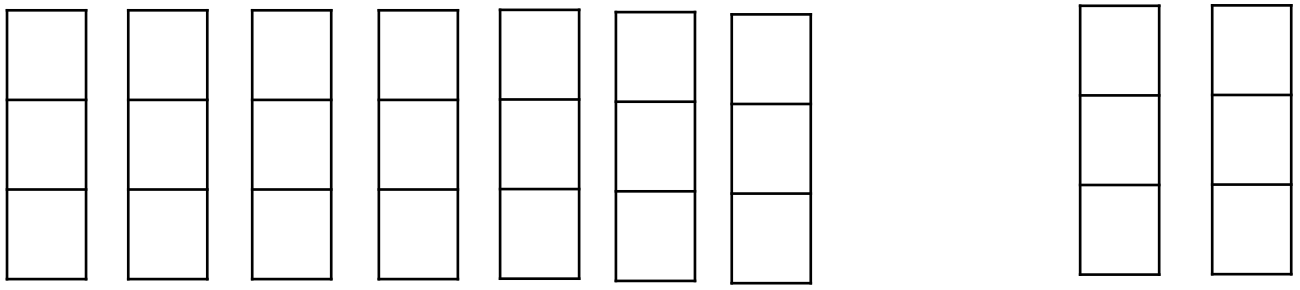
0	1	2	3	0*	1*	4	0	1	2	3	4
0	0	0	0			4	4	4	4	3	3
	1	1	1			1	0	0	0	0	4
		2	2			2	2	1	1	1	1
			3			3	3	3	2	2	2

No of Page Faults: 10
No of frames 4

First In First Out (FIFO) (continue)

Reference String:

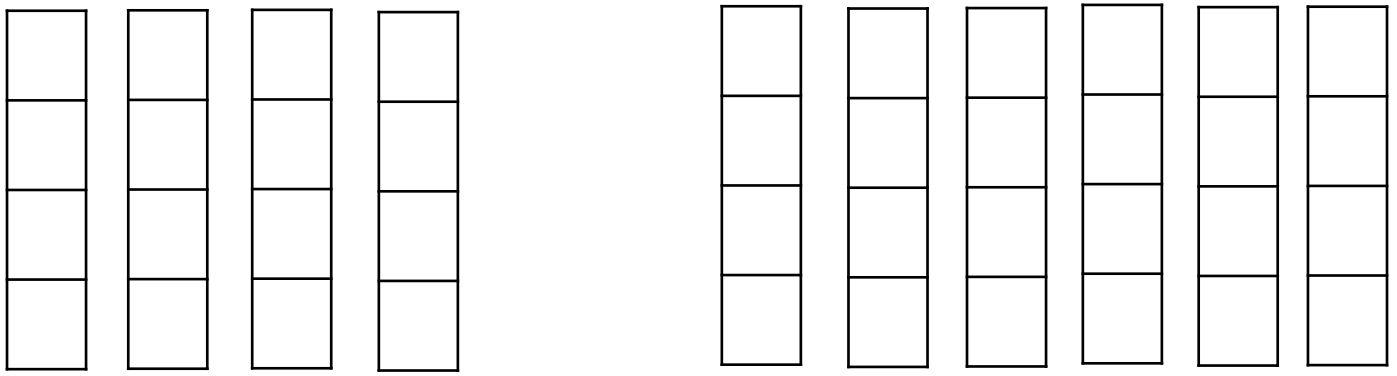
1 2 3 4 1 2 5 1 2 3 4 5



No of Page Faults
No of frames 3

Reference String:

1 2 3 4 1 2 5 1 2 3 4 5



No of Page Faults
No of frames 4

Problem

Problem 1: Consider the virtual page reference string: 1, 2, 3, 2, 4, 1, 3, 2, 4, 1
On a demand paged virtual memory system a main memory of size 3-page frames which are initially empty. Let LRU, FIFO and OPTIMAL denote the number of page faults under the corresponding page replacement policy. Then (Give explanation)

(A) OPTIMAL < LRU < FIFO
(B) OPTIMAL < FIFO < LRU
(C) OPTIMAL = LRU
(D) OPTIMAL = FIFO

Problem 2: A system uses 3 page frames for storing process pages in main memory. Assume that all the page frames are initially empty. What is the total number of page faults that will occur while processing the page reference string given below-

4 , 7, 6, 1, 7, 6, 1, 2, 7, 2

Also calculate the hit ratio and miss ratio if it uses the FIFO, LRU and Optimal page replacement policy.

Problem 1: Consider the virtual page reference string:

1, 2, 3, 2, 4, 1, 3, 2, 4, 1

On a demand paged virtual memory system a main memory of size 3-page frames which are initially empty. Let LRU, FIFO and OPTIMAL denote the number of page faults under the corresponding page replacement policy. Then (Give explanation)

(A) OPTIMAL < LRU < FIFO
(B) OPTIMAL < FIFO < LRU
(C) OPTIMAL = LRU
(D) OPTIMAL = FIFO

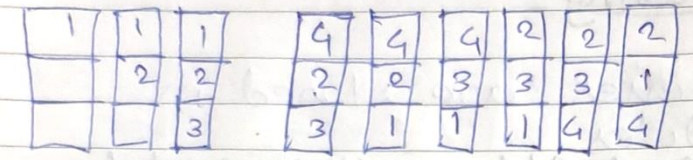
Prob 29 ▸ Monday

Day (180-185)

Week # 27

LRU Page Replacement -

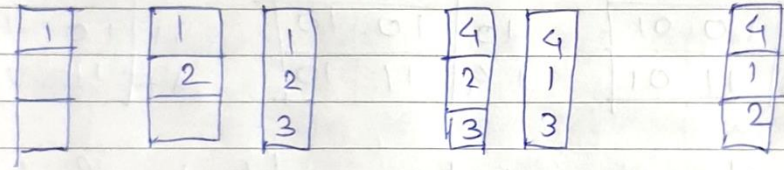
1, 2, 3, 2, 4, 1, 3, 2, 4, 1



⇒ 9 page faults

FIFO -

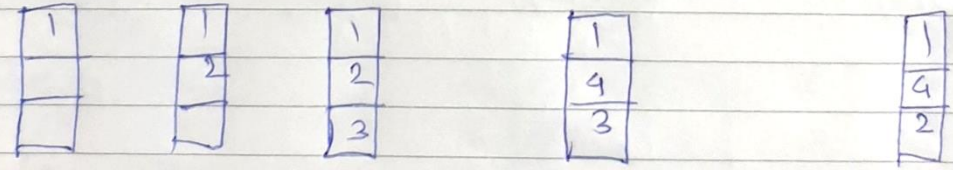
1, 2, 3, 2, 4, 1, 3, 2, 4, 1



⇒ 6 Page faults

Optimal

1, 2, 3, 2, 4, 1, 3, 2, 4, 1



⇒ 5 Page fault.

Optimal < FIFO < LRU

Problem 2: A system uses 3 page frames for storing process pages in main memory. Assume that all the page frames are initially empty. What is the total number of page faults that will occur while processing
the page reference string given below-

4 , 7, 6, 1, 7, 6, 1, 2, 7, 2

Also calculate the hit ratio and miss ratio if it uses the FIFO, LRU and Optimal page replacement policy.

Prob

FIFO

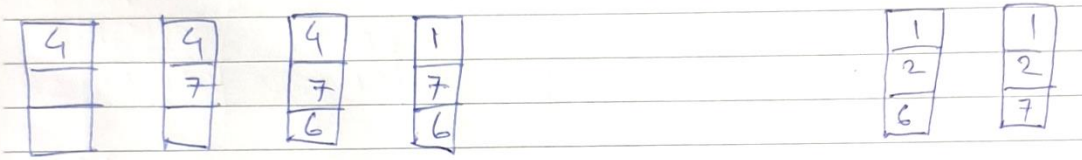
4 , 7, 6, 1, 7, 6, 1, 2, 7, 2



Page Fault = 6, Hit ratio = 4/10 x 100 = 40 %. miss ratio = 60 %.

LRU

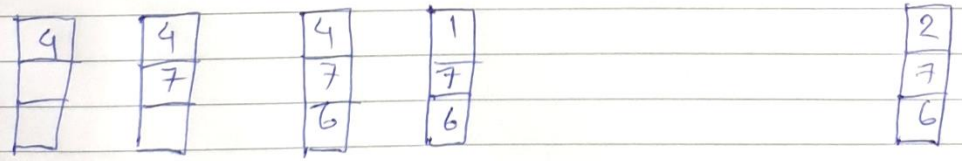
4, 7, 6, 1, 7, 6, 1, 2, 7, 2



Same as FIFO.

Optimal

4, 7, 6, 1, 7, 6, 1, 2, 7, 2



Page fault = 5, Hit ratio = 5/10 x 100 % = 50 %.

Miss ratio = 5/10 x 100 %.
= 50 %

July 2015
S M T W T F S
1 2 3 4
5 6 7 8 9 10 11